# Large Language Models meet Collaborative Filtering: An Efficient All-round LLM-based Recommender System


Sein Kim*
rlatpdlsgns@kaist.ac.kr
KAIST
Republic of Korea

Hongseok Kang*
ghdtjr0311@kaist.ac.kr
KAIST
Republic of Korea

Seungyoon Choi
csyoon08@kaist.ac.kr
KAIST
Republic of Korea

Donghyun Kim
amandus.kim@navercorp.com
NAVER Corporation
Republic of Korea

Minchul Yang
minchul.yang@navercorp.com
NAVER Corporation
Republic of Korea

Chanyoung Park†
cy.park@kaist.ac.kr
KAIST
Republic of Korea





## ABSTRACT

Collaborative filtering recommender systems (CF-RecSys) have shown successive results in enhancing the user experience on social media and e-commerce platforms. However, as CF-RecSys struggles under cold scenarios with sparse user-item interactions, recent strategies have focused on leveraging modality information of user/items (e.g., text or images) based on pre-trained modality encoders and Large Language Models (LLMs). Despite their effectiveness under cold scenarios, we observe that they underperform simple traditional collaborative filtering models under warm scenarios due to the lack of collaborative knowledge. In this work, we propose an efficient **A**ll-round **LLM**-based **Rec**ommender system, called A-LLMRec, that excels not only in the cold scenario but also in the warm scenario. Our main idea is to enable an LLM to directly leverage the collaborative knowledge contained in a pre-trained state-of-the-art CF-RecSys so that the emergent ability of the LLM as well as the high-quality user/item embeddings that are already trained by the state-of-the-art CF-RecSys can be jointly exploited. This approach yields two advantages: (1) model-agnostic, allowing for integration with various existing CF-RecSys, and (2) efficiency, eliminating the extensive fine-tuning typically required for LLM-based recommenders. Our extensive experiments on various real-world datasets demonstrate the superiority of A-LLMRec in various scenarios, including cold/warm, few-shot, cold user, and cross-domain scenarios. Beyond the recommendation task, we also show the potential of A-LLMRec in generating natural language outputs based on the understanding of the collaborative knowledge by performing a favorite genre prediction task. Our code is available at https://github.com/ghdtjr/A-LLMRec.


*Both authors contributed equally to this research.
†Corresponding author.




*KDD '24, August 25–29, 2024, Barcelona, Spain.*





ACM ISBN 979-8-4007-0490-1/24/08
https://doi.org/10.1145/XXXXXX.XXXXXX



## KEYWORDS

Recommender System, Large Language Models, Collaborative Filtering

**ACM Reference Format:**
Sein Kim, Hongseok Kang, Seungyoon Choi, Donghyun Kim, Minchul Yang, and Chanyoung Park. 2024. Large Language Models meet Collaborative Filtering: An Efficient All-round LLM-based Recommender System. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '24), August 25–29, 2024, Barcelona, Spain.* ACM, New York, NY, USA, 12 pages. https://doi.org/10.1145/XXXXXX.XXXXXX


## 1 INTRODUCTION

With the recent exponential growth in the number of users and items, collaborative filtering models [14, 15, 20, 40] encounter the long-standing cold-start problem [1, 43], stemming from the inherent sparsity of user-item interaction data. In other words, for users/items with few interactions, it becomes challenging to construct collaborative knowledge with other similar users/items, leading to suboptimal recommendation performance, especially in the cold-start scenarios. To overcome this issue, recent studies have focused on leveraging modality information of users/items (e.g., user demographics, item titles, descriptions, or images) to enhance recommendation performance under cold-start scenarios. Specifically, MoRec [51] utilizes pre-trained modality encoders (e.g., BERT [9] or Vision-Transformer [10]) to project raw modality features of items (e.g., item texts or images), thereby replacing the item embeddings typically used in collaborative filtering recommendation models. Similarly, CTRL [25] considers tabular data and its textual representation as two different modalities and uses them to pre-train collaborative filtering recommendation models through a contrastive learning objective, which is then fine-tuned for specific recommendation tasks.

Despite the effectiveness of modality-aware recommender systems in cold scenarios, the recent emergence of Large Language Models (LLMs), known for their rich pre-trained knowledge and advanced language understanding capabilities, has attracted significant interest in the recommendation domain to effectively extract

[1]An item is categorized as 'warm' if it falls within the top 35% of interactions, and if it falls within the bottom 35%, it is classified as a 'cold' item.
[2]After training each model using all the available data in the training set, we separately evaluate on cold and warm items in the test set.

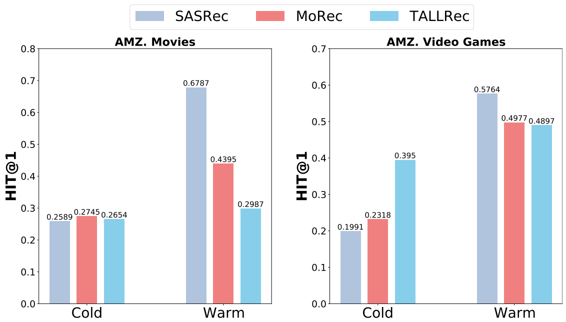


**Figure 1: Comparisons between collaborative filtering model (SASRec), modality-aware model (i.e., MoRec), and LLM-based model (i.e., TALLRec) under the cold/warm[1] scenarios on Amazon Movies/Video Games dataset (Hit@1)[2].**

and integrate modality information [37, 48]. Early studies on LLM-based recommendation [12, 16, 44] have employed OpenAI-GPT with *In-context Learning* [4]. This approach adapts to new tasks or information based on the context provided within the input prompt and demonstrates the potential of LLMs as a recommender system. Moreover, to bridge the gap between the training tasks of LLMs and recommendation tasks, TALLRec [2] fine-tunes LLMs with recommendation data using LoRA [18]. This approach has empirically demonstrated that, in cold scenarios and cross-domain scenarios, fine-tuned LLMs outperform traditional collaborative filtering models.

Although modality-aware and LLM-based recommender systems have proven effective in cold scenarios with limited user-item interactions, we argue that these methods suffer from the lack of collaborative knowledge due to their heavy reliance on textual information [51]. Consequently, *when abundant user-item interactions are available (i.e., warm scenario), modality-aware and LLM-based recommenders are rather inferior to simple traditional collaborative filtering models.* As shown in Figure 1, while the modality-aware recommender (i.e., MoRec) and the LLM-based recommender (i.e., TALLRec) significantly outperform the traditional collaborative filtering model (i.e., SASRec [20]) in the cold scenario, they are outperformed by the traditional collaborative filtering model in the warm scenario. This is mainly because the textual information becomes less important in the warm scenario, where ID-based collaborative filtering models excel at modeling popular items [6, 51]. However, while excelling in the cold scenario is crucial, the majority of user interactions and the revenue are predominantly generated from already existing and active items (i.e., warm items) in real-world application of recommendation systems, which contribute up to 90% of interactions in offline-industrial data [8, 49]. Furthermore, as demonstrated by DCBT [49], modeling both warm and cold items is essential for improving overall user engagement, which is evidenced by A/B testing with real-world industrial data. This implies that the warm scenario should not be overlooked.

In this paper, we propose an efficient all-round LLM-based recommender system, called A-LLMRec (**A**ll-round **LLM**-based **Rec**ommender system), that excels not only in the cold scenario but also in the warm scenario (hence, all-round recommender system). Our main idea is to enable an LLM to directly leverage the collaborative knowledge contained in a pre-trained state-of-the-art collaborative filtering recommender system (CF-RecSys) so that the emergent ability [45] of the LLM, as well as the high-quality user/item embeddings that are already trained by the state-of-the-art CF-RecSys, can be jointly exploited. More precisely, we devise an alignment network that aligns the item embeddings of the CF-RecSys with the token space of the LLM, aiming at transferring the collaborative knowledge learned from a pre-trained CF-RecSys to the LLM enabling it to understand and utilize the collaborative knowledge for the downstream recommendation task.

The key innovation of A-LLMRec is that it *requires the fine-tuning of neither the CF-RecSys nor the LLM*, and that the alignment network is the only neural network that is trained in A-LLMRec, which comes with the following two crucial advantages:

(1) **(Model-agnostic)** A-LLMRec allows any existing CF-RecSys to be integrated, which implies that services using their own recommender models can readily utilize the power of the LLM. Besides, any updates of the recommender models can be easily reflected by simply replacing the old models, which makes the model practical in reality.
(2) **(Efficiency)** A-LLMRec is efficient in that the alignment network is the only trainable neural network, while TALLRec [2] requires the fine-tuning of the LLM with LoRA [18]. As a result, A-LLMRec trains approximately **2.53** times and inferences **1.71** times faster than TALLRec, while also outperforming both TALLRec and CF-RecSys in both cold and warm scenarios.

Our extensive experiments on various real-world datasets demonstrate the superiority of A-LLMRec, revealing that aligning high-quality user/item embeddings with the token space of the LLM is the key for solving not only cold/warm scenarios but also few-shot, cold user, and cross-domain scenarios. Lastly, beyond the recommendation task, we perform a language generation task, i.e., favorite genre prediction, to demonstrate that A-LLMRec can generate natural language outputs based on the understanding of users and items through the aligned collaborative knowledge from CF-RecSys. Our main contributions are summarized as follows:

- We present an LLM-based recommender system, called A-LLMRec, that directly leverages the collaborative knowledge contained in a pre-trained state-of-the-art recommender system.
- A-LLMRec requires the fine-tuning of neither the CF-RecSys nor the LLM, while only requiring an alignment network to be trained to bridge between them.
- Our extensive experiments demonstrate that A-LLMRec outperforms not only the conventional CF-RecSys in the warm scenario but also the LLMs in the cold scenario.

## 2 RELATED WORK

### 2.1 Collaborative Filtering

Collaborative Filtering (CF) is the cornerstone of recommendation systems, fundamentally relying on leveraging users' historical preferences to inform future suggestions. The key idea is to rely on similar users/items for recommendations. The emergence of matrix factorization marked a significant advancement in CF, as evidenced by numerous studies [19, 22, 38], demonstrating its superiority in capturing the latent factors underlying user preferences.

This evolution continued with the introduction of Probabilistic Matrix Factorization (PMF) [5, 33] and Singular Value Decomposition (SVD) [30, 53], which integrate probabilistic and decomposition techniques to further refine the predictive capabilities of CF models. AutoRec [39] and Neural Matrix Factorization (NMF) [15] utilized deep learning to enhance CF by capturing complex user-item interaction patterns. Recently, [7, 21, 34, 36] proposed modeling collaborative filtering based on sequential interaction history. Caser [41] and NextItNet [50] utilize Convolutional Neural Networks (CNNs) [23] to capture the local sequence information, treating an item sequence as images. While these methods effectively capture user preferences using interaction history, including user and item IDs, they overlook the potential of the modality information of the user/item, which could enhance model performance and offer a deeper analysis of user behaviors.

### 2.2 Modality-aware Recommender Systems

Modality-aware recommenders utilize modality information such as item titles, descriptions, or images to enhance the recommendation performance mainly under cold scenarios. Initially, CNNs were used to extract visual features, modeling human visual preferences based on Mahalanobis distance [31]. With advancements in pre-trained modality encoders like BERT [9, 27, 29, 47, 51] and ResNet/Vision-Transformer [10, 11], modality-aware recommender systems have accelerated research by utilizing modality knowledge on recommendation tasks. For example, NOVA [27] and DMRL [28] proposed non-invasive fusion and disentangled fusion of modality, respectively, by carefully integrating pure item embeddings and text-integrated item embeddings using the attention mechanism. MoRec [51] leverages modality encoders to project raw modality features, thereby replacing item embeddings used in collaborative filtering models. As for the pre-training based models, Liu et al. [29] constructs user-user and item-item co-interaction graphs to extract collaborative knowledge, then integrates with user/item text information through attention mechanism in an auto-regressive manner, and CTRL [25] pre-trains the collaborative filtering models using paired tabular data and textual data through a contrastive learning objective, subsequently fine-tuning them for recommendation tasks. Most recently, RECFORMER [24] proposed to model user preferences and item features as language representations based on the Transformer architecture by formulating the sequential recommendation task as the next item sentence prediction task, where the item key-value attributes are flattened into a sentence.

### 2.3 LLM-based Recommender Systems

Recently, research on LLMs has gained prominence in the field of modality-aware recommendation systems, with LLM-based recommendations emerging as a significant area of focus. The pre-trained knowledge and the reasoning power of LLMs based on the advanced comprehension of language are shown to be effective for recommendation tasks, and many approaches have been proposed leveraging LLM as a recommender system. More precisely, [12, 16, 44] utilize LLMs with *In-context Learning* [4], adapting to new tasks or information based on the context provided within the input prompt. For example, Sanner et al. [37] employs In-context Learning for recommendation tasks, exploring various prompting styles such as completion, instructions, and few-shot prompts based on item texts and user descriptions. Gao et al. [12] assigns the role of a recommender expert to rank items that meet users' needs through prompting and conducts zero-shot recommendations. These studies empirically demonstrated the potential of LLMs using its rich item information and natural language understanding in the recommendation domain. However, these approaches often underperform traditional recommendation models [20, 40], due to the gap between the natural language downstream tasks used for training LLMs and the recommendation task [2]. To bridge this gap, TALLRec [2] employs the *Parameter Efficient Fine-Tuning* (PEFT) method, also known as LoRA [18]. This methodology enables TALLRec to demonstrate enhanced efficacy, surpassing traditional collaborative filtering recommendation models, particularly in mitigating the challenges posed by the cold start dilemma and in navigating the complexities of cross-domain recommendation scenarios. However, it is important to note that since TALLRec simply converts the conventional recommendation task into an instruction text and uses it for fine-tuning, it still fails to explicitly capture the collaborative knowledge that is crucial in warm scenarios.

## 3 PROBLEM FORMULATION

In this section, we introduce a formal definition of the problem including the notations and the task description.

**Notations.** Let $\mathcal{D}$ denote the historical user-item interaction dataset $(\mathcal{U}, \mathcal{I}, \mathcal{T}, \mathcal{S}) \in \mathcal{D}$, where $\mathcal{U}, \mathcal{I}, \mathcal{T}$, and $\mathcal{S}$ denote the set of users, items, item titles/descriptions, and item sequences, respectively. $\mathcal{S}^u = (i_1^u, i_2^u, \cdots, i_k^u, \cdots i_{|\mathcal{S}^u|}^u) \in \mathcal{S}$ is a sequence of item interactions of a user $u \in \mathcal{U}$, where $i_k^u$ denotes the $k$-th interaction of user $u$, and this corresponds to the index of the interacted item in the item set $\mathcal{I}$. Moreover, each item $i \in \mathcal{I}$ is associated with title and description text $(t^i, d^i) \in \mathcal{T}$.

**Task: Sequential Recommendation.** The goal of sequential recommendation is to predict the next item to be interacted with by a user based on the user's historical interaction sequence. Given a set of user historical interaction sequences $\mathcal{S} = \left\{ \mathcal{S}^1, \mathcal{S}^2, \cdots, \mathcal{S}^{|\mathcal{U}|} \right\}$, where $\mathcal{S}^u$ denotes the sequence of user $u$, the subset $\mathcal{S}_{1:k}^u \subseteq \mathcal{S}^u$ represents the sequence of user $u$ from the first to the $k$-th item denoted as $\mathcal{S}_{1:k}^u = (i_1^u, i_2^u, \cdots, i_k^u)$. Given an item embedding matrix $\mathbf{E} \in \mathbb{R}^{|\mathcal{I}| \times d}$, the embedding matrix of items in $\mathcal{S}_{1:k}^u$ is denoted by $\mathbf{E}_{1:k}^u = (\mathbf{E}_{i_1^u}, \mathbf{E}_{i_2^u}, ..., \mathbf{E}_{i_k^u}) \in \mathbb{R}^{k \times d}$, where $\mathbf{E}_{i_j^u}$ denotes the $i_j^u$-th row of $\mathbf{E}$. This sequence embedding matrix is fed into a collaborative filtering recommender (e.g., SASRec [20]) to learn and predict the next item in the user behavior sequence $\mathcal{S}_{1:k}^u$ as follows:

$$\max_{\Theta} \prod_{u \in \mathcal{U}} \prod_{k=1}^{|\mathcal{S}^u|-1} p(i_{k+1}^u | \mathcal{S}_{1:k}^u; \Theta) \tag{1}$$

where $p(i_{k+1}^u | \mathcal{S}_{1:k}^u; \Theta)$ represents the probability of the $(k+1)$-th interaction of user $u$ conditioned on the user's historical interaction sequence $\mathcal{S}_{1:k}^u$, and $\Theta$ denotes the set of learnable parameters of the collaborative filtering recommender (CF-RecSys). By optimizing $\Theta$ to maximize Equation 1, the model can obtain the probability of the next items for user $u$, over all possible items.

It is important to note that although we mainly focus on the sequential recommendation task in this work, A-LLMRec can also be